Beyond Lesion Burden: Circulating PINK1/Parkin Distinguishes Vascular Parkinsonism and Predicts Incident Parkinsonism in Cerebral Small-Vessel Disease

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ABSTRACT

BACKGROUND: Vascular parkinsonism (VP) develops in only a subset of patients with cerebral small-vessel disease (CSVD), and lesion burden alone does not explain susceptibility.

OBJECTIVE: Test whether circulating PINK1 and Parkin separate VP from lesion-matched CSVD and predict incident parkinsonism.

METHODS: 118 VP patients, 118 CSVD controls matched on WMH decile and lacune count, and 92 idiopathic PD patients. Serum biomarkers were measured by blinded ELISA. Imaging was adjudicated blinded.

RESULTS: PINK1 and Parkin were highest in VP. Adding both markers raised AUC from 0.75 to 0.84. In 452 CSVD patients, baseline PINK1 predicted 30-month progression.

CONCLUSION: PINK1/Parkin biology may explain parkinsonian susceptibility beyond visible CSVD lesion burden.

OBJECTIVES

STUDY QUESTION:
Can mitophagy markers explain why similar CSVD lesion burden produces VP in some patients but not others?

OBJECTIVES:
- Compare PINK1/Parkin across VP, lesion-matched CSVD, and idiopathic PD.
- Quantify added discrimination beyond clinical and imaging variables.
- Test whether baseline PINK1 predicts incident parkinsonism.
- Link hypoperfusion, PINK1 knockdown, white-matter injury, and gait.

HYPOTHESIS:
A higher PINK1/Parkin signal identifies a biologically vulnerable CSVD subgroup.

METHODS

DESIGN:
Prospective biomarker-imaging study with lesion-matched controls and an independent follow-up cohort.

COHORTS (TABLE 1):

Group	n	Role
VP	118	Target phenotype
Lesion-matched CSVD	118	Lesion-matched control
Idiopathic PD	92	Neurodegenerative comparator
CSVD follow-up	452	Risk cohort, followed 30 months

MEASURES:
Blinded serum PINK1/Parkin ELISA; WMH volume, Fazekas grade, lacunes, and strategic basal-ganglia infarcts.

MODELS:
Adjusted for demographic, vascular, renal, metabolic, and dopaminergic-treatment covariates; bootstrap optimism correction; competing-risk analysis of incident parkinsonism.

MECHANISM:
Chronic hypoperfusion tested serum-brain concordance, mt-Keima mitophagic flux, and PINK1 knockdown.

RESULTS 1: SERUM BIOMARKERS BY DIAGNOSTIC GROUP

BIOMARKER SEPARATION: PINK1: VP 53.8±10.4 vs CSVD 43.7±9.8 vs PD 39.6±9.5 pg/mL. Parkin: VP 71.8±11.5 vs CSVD 59.6±10.7 vs PD 51.2±10.3 pg/mL. Both markers: P<0.001 across groups.
INTERPRETATION: The signal persists after matching on WMH and lacunes, so it is not just a lesion-volume proxy.

PINK1: 53.8 vs 43.7 vs 39.6 pg/mL; P<0.001 | **Parkin: 71.8 vs 59.6 vs 51.2 pg/mL; P<0.001**

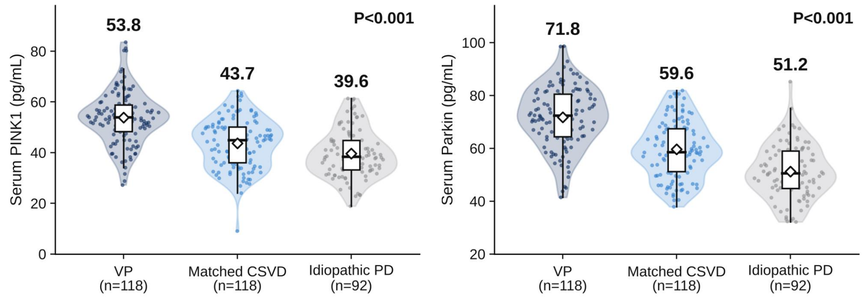
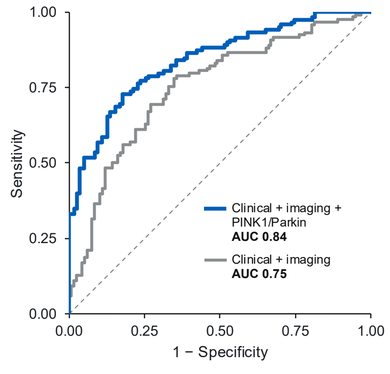


FIGURE 1. SERUM MITOPHAGY MARKERS. Violin, distribution; box, median and IQR; diamond and number, group mean; points, individual participants. Matched CSVD, lesion-matched CSVD controls. P values, overall three-group comparison.

RESULTS 2: DISCRIMINATION AND ADJUSTED ASSOCIATION

DISCRIMINATION: Clinical plus imaging AUC 0.75; + PINK1/Parkin AUC 0.84 (ΔAUC 0.09, P=0.004; optimism-corrected 0.82).
INDEPENDENT ASSOCIATION WITH VP: PINK1 OR 2.08 per SD (95% CI 1.49–2.91); Parkin OR 1.76 (95% CI 1.28–2.42).



AUC, VP VS LESION-MATCHED CSVD
0.75 → 0.84
+0.09, P=0.004
optimism-corrected AUC 0.82

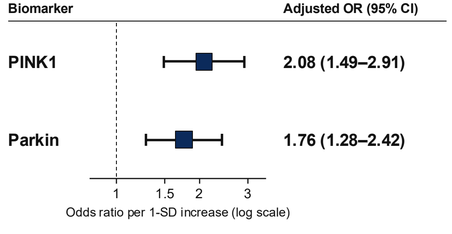
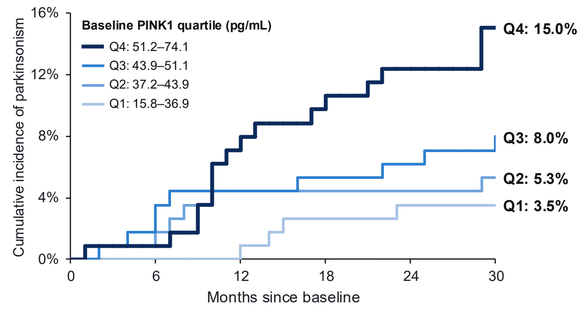


FIGURE 2. MODEL GAIN AND ADJUSTED ASSOCIATION. Left, ROC curves for VP vs lesion-matched CSVD (n=236); AUC labels are the reported values. Right, adjusted odds ratios per 1-SD increase with 95% CI.

RESULTS 3: PROSPECTIVE CSVD COHORT (n=452)

PROSPECTIVE RISK: 36/452 CSVD participants developed parkinsonian signs over 30 months. Baseline PINK1 predicted progression: HR 1.74 per SD, 95% CI 1.21 to 2.49.



36/452 incident cases; adjusted HR 1.74 per SD (95% CI 1.21–2.49)

FIGURE 3. INCIDENT PARKINSONISM BY BASELINE PINK1 QUARTILE. Cumulative incidence with death as a competing risk; n=113 per quartile; events Q1 to Q4: 4, 6, 9, 17.

RESULTS 4: MECHANISTIC SUPPORT

MECHANISTIC SUPPORT: Hypoperfusion serum-brain PINK1 correlation: r=0.60, P<0.001. PINK1 knockdown reduced mitophagic flux and worsened white-matter injury and gait, despite similar CBF in the two hypoperfusion groups.

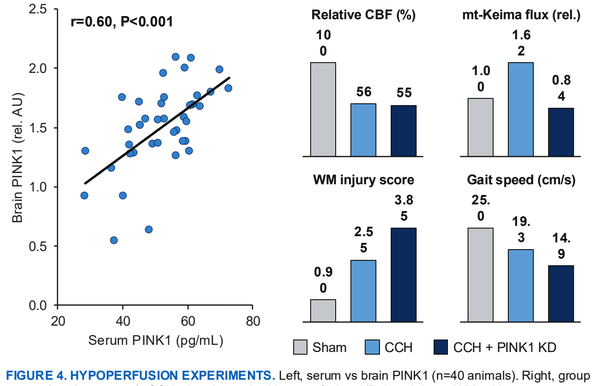


FIGURE 4. HYPOPERFUSION EXPERIMENTS. Left, serum vs brain PINK1 (n=40 animals). Right, group means (n=12 per group). CCH, chronic cerebral hypoperfusion; KD, knockdown; WM, white matter. Knockdown effect sizes are illustrative.

DISCUSSION

KEY MESSAGE:
PINK1/Parkin identifies a biologically vulnerable CSVD subgroup rather than simply reflecting lesion burden.

WHY IT MATTERS:
- VP is difficult to separate from idiopathic PD and non-parkinsonian CSVD by imaging alone.
- A blood marker panel could support risk stratification, trial enrichment, and mechanistic phenotyping.

LIMITATIONS:
- VP can overlap with mixed vascular and neurodegenerative pathology.
- Serum markers may reflect systemic stress, not only brain mitophagy.
- External validation and assay harmonization are needed.

CONCLUSIONS

- Circulating PINK1 and Parkin identify a biological dimension of vascular parkinsonism not captured by conventional lesion burden.
- PINK1 and Parkin distinguish VP from lesion-matched CSVD and idiopathic PD.
- Biomarkers add discrimination beyond clinical and imaging variables.
- In prospective CSVD follow-up, PINK1 additionally predicted incident parkinsonism.
- Hypoperfusion experiments supported concordance with PINK1-dependent mitophagy.

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STRUCTURAL LESION BURDEN IDENTIFIES THE VASCULAR INJURY. PINK1/PARKIN MAY HELP IDENTIFY WHO BECOMES PARKINSONIAN.

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